COMMENSURATED SUBGROUPS OF BRANCH GROUPS: THE MISSING JUST-INFINITE HYPOTHESIS AND HANOI GROUP COUNTEREXAMPLE

ARGUS OPEN RESEARCH PROJECT

Abstract. Wesolek (2016, Theorem 1.1) proved that every commensurated subgroup of a finitely generated *just-infinite* branch group is either finite or of finite index. When the just-infinite hypothesis is omitted, the literal statement fails: we certify that in the finitely generated, faithful, level-transitive 3-peg Hanoi towers branch group $G = H^{(3)}$ acting on the ternary rooted tree, the second derived subgroup $H = G''$ is normal (hence commensurated with $[H : H \cap g^{-1}Hg] = 1$ for all $g \in G$), infinite, and of infinite index in G.

1. Main Counterexample

Let $G = H^{(3)} = \langle a, b, c \rangle \leq \mathrm{Aut}(\mathcal{T}_3)$ be the 3-peg Hanoi group (Bartholdi, Siegenthaler, and Zalesskii, 2010), where the rooted automorphisms satisfy

$$a = (1\,2)(1, 1, a), \qquad b = (1\,3)(1, b, 1), \qquad c = (2\,3)(c, 1, 1).$$

Theorem 1.1. *Let $G = H^{(3)}$ and let $H = G'' = [[G, G], [G, G]]$ be its second derived subgroup. Then:*

(1) *G is a finitely generated, faithful, level-transitive branch group on the ternary tree $\mathcal{T}_3$.*

(2) *Because $H = G''$ is characteristic (hence normal) in G, $g^{-1}Hg \cap H = H$ has index 1 in H for every $g \in G$, so H is a commensurated subgroup of G.*

(3) *$H = G''$ is infinite: the rigid stabilizer containments $\mathrm{rist}_G(u) \geq G'$ and the self-similar embedding of disjoint subtree copies produce infinitely many distinct nontrivial elements in G''.*

(4) *$[G : G''] = \infty$: from the recursive presentation of G, the relation matrix of the metabelian quotient G'/G'' has integer Smith normal form diagonal $(1, 3, 0, 0, 0)$, giving*

$$G'/G'' \;\cong\; \mathbb{Z}^3 \times C_3,$$

and since $[G : G'] = 8$, G'' has infinite index in G.

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